BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 124 · Testing and Test Data

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does anticipating misuse mean in defensive design?

Q2. Merge sorted [1, 5] and [4, 6]. Which result is ascending?
A. [1, 5, 4, 6]
B. [4, 6, 1, 5]
C. [6, 5, 4, 1]
D. [1, 4, 5, 6]

Q3. “A comparison operator must always produce a text paragraph.” Correct this. Explain why.

Q4. A school shares files and printers on one site. “A LAN must span several countries.” Correct this. Why does a LAN fit here?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Normal test — A typical value within permitted range.

Boundary test — A value at the smallest or largest allowed limit.

Invalid test — A value of the right type but outside allowed rules.

Erroneous test — Input of the wrong type that the program should reject safely.

New learning

Normal test: A typical value within permitted range.

Boundary test: A value at the smallest or largest allowed limit.

Invalid test: A value of the right type but outside allowed rules.

Erroneous test: Input of the wrong type that the program should reject safely.

Worked example

Create a test table for an integer score from 0 to 100 inclusive.

1. Normal 50 should be accepted.

2. Boundary 0 and 100 should both be accepted.

3. Invalid -1 and 101 should both be rejected.

4. Erroneous 'hello' should produce a safe error, not be accepted.

We Do · Practise with guidance

Guided 1

Plan tests for a quantity between 1 and 10.

Worked method: An erroneous text input is 'ten'; expected outcome is rejection.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Choose a negative test value to check that the program rejects it.

```
fileMB = 700
if fileMB >= 0 AND fileMB <= 1000 then
    print("Accepted")
else
    print("Rejected")
endif
```

test

Question 1

Identify two boundary tests for valid ages 13–18. [2 marks]

Question 2

Distinguish invalid and erroneous data for a numeric mark. [4 marks]

Question 3

State why expected outcomes belong in a test plan. [2 marks]

Exit Ticket · Answer before checking

For range 1–10, 11 is a valid boundary value.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Planning for invalid, unexpected or malicious use and handling it safely.

Q2. D — [1, 4, 5, 6]

Q3. A Boolean result: true or false.

Q4. A LAN connects devices in a small area, such as one school site. A WAN links networks over a large area. Internet access alone does not make a LAN a WAN.

I Do / We Do pages · Guided practice

Q1. An erroneous text input is 'ten'; expected outcome is rejection.

You Do pages · Code First

```
fileMB = -1
if fileMB >= 0 AND fileMB <= 1000 then
    print("Accepted")
else
    print("Rejected")
endif
```

Expected result: Rejected

You Do pages · Independent questions

Q1. 13 and 18.

Q2. Invalid: numeric 101 outside 0–100; erroneous: text 'apple' where an integer is expected.

Q3. They allow actual results to be compared with requirements and faults detected.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: